

Smart Glasses Just Got Smarter

Golden-i Gen 3.8 is a handsfree wireless computing and communications headset that operates by voice commands and head movements. It was demonstrated at Electronics Rocks Conference 2013

ANAGHA P.

We all have seen the media frenzy that surrounded Google's Glass, a wearable computer that integrated a lot of Android functionality, while making you look like an Android at the same time.

Golden-i is a mobile wireless wearable headset computer with 38cm (15-inch) near-eye virtual display, which enables the users to carry out general computer functions without using their hands. It allows the users to access six independent devices or networks simultaneously (including PCs, cellular phones and industrial and enterprise servers). The device supports more than 38 languages. It is operated by voice-user interface and head-tracking system.

Technology

The device has an embedded 14MP camera that can record and/or send full-colour images or streaming video in real time, in resolutions ranging from 240p to 1080p, at the rate of 2fps to 60fps. In addition, it is equipped with a head tracker, Wi-Fi, Bluetooth, ambient-light sensor, USB on-the-go and other peripherals. The design also includes a low-power, ultra-small form factor micro-display that integrates advanced voice and gesture recognition technology.

The latest version of Golden-i, the Gen 3.8 system, is built around TI's powerful OMAP4 processor and runs on Android 4.2 operating system. A Windows Embedded Compact 7 version of the product is also supported. An automated test suite is available for hardware design validation and manufacturing test.



Construction worker using Golden-i

Applications

Golden-i is a finished product and is being adopted in various fields, such as fire brigade, police department, construction, maintenance, repair and general professional applications. It is used in healthcare by doctors and nurses, and for emergency medical services. Product developers can also use Golden-i as a development platform.

Since this gadget is highly customisable, it can be used in a variety of applications, and robust solutions and support are provided to the clients by the firm.

Here is an example of how Golden-i Gen 3.8 is used in a fire brigade. The application used here is the Firefighter Pro, which helps improve time efficiency and safety. It allows firefighters to access vital information while dealing with an emergency situation. They could call up floor plans and GPS coordinates on their wearable computer, giving them real-time understanding of the layout, assisting navigation through unknown struc-

tures efficiently (using assisted-GPS inertial navigation) and hence enabling quick evacuation/rescue operations. It helps the firefighter see through smoke, mist, dust and thin materials using an optional IR camera.

The firefighters can monitor the vital signs of individuals and the crew (which is an ap-

plication in healthcare field too). Live, on-site video streaming is also possible to and from the affected environment.

Development

The Golden-i is a global multinational endeavour that was developed with the goal of improving productivity, efficiency and safety in all professional and industrial sectors. It supports and guides the user at any moment without interruption.

The US-based Kopin Corporation's portfolio includes designing, building and deploying lightweight, power-efficient, ultra-small liquid crystal displays. They entered into a partnership with Bangalore-based Mistral Solutions to design the headset computing and communications system. Mistral also manufactures compact hardware electronics required by Golden-i.

The concept of handsfree, wireless, headset computer was first discussed in August 2006. The first actual Golden-i headset development started on July 1, 2007, and in January 2008, the